This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-GA-1201A-01** Sheet 1 of 1

OPL Title	**Mechanical Seal Flush (API Plan 11) Verification**	**Discipline**	Mechanical
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **mechanical seal flush (api plan 11) verification** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Confirm seal flush dP > 1.5 bar before start. Low flush = seal face overheating and hexane leak to atmosphere. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Line up discharge flush to seal chamber and open flush isolation	Isolation & LOTO verified
2	Check PDI-1201 reads 1.5-2.5 bar dP across the flush orifice RO-1201	Parts match BOM / spec
3	Verify no visible leak or drip at the gland / seal drain	Torque / clearance to spec
4	If dP low, isolate and inspect orifice RO-1201 for plugging by polymer/dirt	No leak / smooth operation
5	Confirm quench (Plan 62) N2/steam is lined up to the outboard side	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr.	API Plan 11 flush orifice RO-1201 partially plugged causing seal face.	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v.
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to.
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Confirm seal flush dP > 1.5 bar before start. Low flush = seal face overheating and hexane leak to atmosphere.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Sunday, 22 March 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only

ONE POINT LESSON (OPL)

OPL No: **OPL-GA-1201A-02**
Sheet 1 of 1

OPL Title	**Bearing Oil Bath Level & Greasing**	**Discipline**	Mechanical
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **bearing oil bath level & greasing** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Maintain oil bath at the centre of the sight glass with ISO VG 68. Both over-fill and under-fill overheat the bearing. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Stop pump, isolate electrically (LOTO) before any lube work	Isolation & LOTO verified
2	Check the oil sight-glass level sits at the middle mark	Parts match BOM / spec
3	Top-up with ISO VG 68 only - never mix grades	Torque / clearance to spec
4	Inspect the constant-level oiler bottle and refill	No leak / smooth operation
5	Record oil condition (colour/water) on the weekly lube route sheet	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr.	API Plan 11 flush orifice RO-1201 partially plugged causing seal face.	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v.
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to.
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Maintain oil bath at the centre of the sight glass with ISO VG 68. Both over-fill and under-fill overheat the bearing.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Thursday, 26 March 2026

Notes: Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-GA-1201A-03** Sheet 1 of 1

OPL Title	**Pump-Motor Alignment Check (Laser)**	**Discipline**	Mechanical
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **pump-motor alignment check (laser)** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Shaft misalignment is the #1 root cause of GA-1201A high-vibration trips (VSHH-1201) and premature bearing/seal failure. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and LOTO the motor, remove coupling guard	Isolation & LOTO verified
2	Mount laser alignment brackets across the coupling spacer	Parts match BOM / spec
3	Rotate 0-90-180-270 deg and capture readings	Torque / clearance to spec
4	Target angular < 0.05 mm/100 mm and offset < 0.05 mm	No leak / smooth operation
5	Add/remove shims under motor feet, re-check, and record the alignment report	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr.	API Plan 11 flush orifice RO-1201 partially plugged causing seal face.	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v.
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to.
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Shaft misalignment is the #1 root cause of GA-1201A high-vibration trips (VSHH-1201) and premature bearing/seal failure.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Monday, 30 March 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-GA-1201A-04** Sheet 1 of 1

OPL Title	**Minimum Flow Line Operation & Deadhead Protection**	**Discipline**	Process / Operations
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **minimum flow line operation & deadhead protection** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Never run the pump against a closed discharge. The min-flow recycle (FV-1201) protects the pump from deadhead overheating. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm FV-1201 min-flow recycle line is open before start	Permit valid & briefed
2	Watch that FIC-1201 stays above the 9 m3/h minimum continuous flow	Parameter within limit
3	If FSLL-1201 trips, check downstream XV-1201 and manual valves for a closed path	Interlock status confirmed
4	Do not defeat the min-flow interlock to force higher discharge pressure	Change logged on DCS
5	Reset the pump only after the low-flow cause has been cleared	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr	API Plan 11 flush orifice RO-1201 partially plugged causing seal face	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Never run the pump against a closed discharge. The min-flow recycle (FV-1201) protects the pump from deadhead overheating.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Friday, 03 April 2026

Notes: Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-GA-1201A-05** Sheet 1 of 1

OPL Title	**Cold Alignment vs Hot Check for Hexane Service**	**Discipline**	Mechanical
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **cold alignment vs hot check for hexane service** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Hexane at operating temperature grows the casing; a good cold alignment must include the correct thermal-growth offset. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Take cold alignment with the specified target offsets applied	Isolation & LOTO verified
2	After ~2 h at temperature, do a hot-check on the coupling	Parts match BOM / spec
3	Compare hot readings against the cold targets	Torque / clearance to spec
4	Adjust thermal-growth compensation if repeat vibration trips occur	No leak / smooth operation
5	Log both cold and hot data in the machine history file	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr.	API Plan 11 flush orifice RO-1201 partially plugged causing seal face.	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v.
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to.
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Hexane at operating temperature grows the casing; a good cold alignment must include the correct thermal-growth offset.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Tuesday, 07 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID;, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only

ONE POINT LESSON (OPL)

OPL No: **OPL-GA-1201A-06**
Sheet 1 of 1

OPL Title	**Start-Up & Priming Procedure GA-1201A**	**Discipline**	Process / Operations
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **start-up & priming procedure ga-1201a** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Correct priming and venting prevents a dry-run that instantly destroys the mechanical seal. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Open the suction valve fully and vent the casing air via N5 vent	Permit valid & briefed
2	Establish seal flush Plan 11 and confirm PDI-1201 dP	Parameter within limit
3	Confirm the min-flow recycle line FV-1201 is open	Interlock status confirmed
4	Bump-start the motor and check correct rotation direction	Change logged on DCS
5	Start the motor and confirm PIC-1201 pressure and vibration settle within 5 s	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Hexane leak observed at GA-1201A seal gland during operation, seal dr	API Plan 11 flush orifice RO-1201 partially plugged causing seal face	Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v
GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s	Angular misalignment 0.12 mm/100mm after foundation settlement	Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to
Bearing DE noisy with rising temperature TI-1201 trend	Outer race spalling on DE bearing due to prolonged misalignment	Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68

6. KEY LEARNING POINTS

- Correct priming and venting prevents a dry-run that instantly destroys the mechanical seal.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Saturday, 11 April 2026

Notes: Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-GA-1201A-07** Sheet 1 of 1

OPL Title	**Vibration Trend Monitoring & Alarm Response (VSHH-1201)**	**Discipline**	Instrument
Equipment	GA-1201A - HEXANE FEED PUMP	**Area / Unit**	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-1201

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **vibration trend monitoring & alarm response (vshh-1201)** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. A rising vibration trend predicts bearing failure days in advance and lets you plan a repair before a trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Record VT-1201 on the daily PdM route in mm/s RMS	Loop isolated / under permit
2	Note alarm at 4.5 mm/s and trip at 7.1 mm/s (VSHH-1201)	Reference standard traceable
3	A rise greater than 1 mm/s within a week = plan a bearing inspection	As-found & as-left recorded
4	Correlate the vibration rise with TI-1201 bearing temperature	Trip / control action verified
5	Escalate to reliability engineer if the 1x running-speed component dominates	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled proof test of low-low suction pressure trip	Switch set-point drift found +0.08 barg during proof test	Recalibrated PSLL-1201 to 0.5 barg trip, function tested to DCS
PDI-1201 reading erratic, seal-flush low alarm nuisance-tripping	Impulse lines fouled with hexane residue/polymer fines	Cleaned impulse lines and re-zeroed PDI-1201 transmitter
Discharge flow control hunting, FIC-1201 oscillating	Loop poorly tuned after transmitter re-range	Re-ranged FT-1201 and retuned FIC-1201 PID parameters

6. KEY LEARNING POINTS

- A rising vibration trend predicts bearing failure days in advance and lets you plan a repair before a trip.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Wednesday, 15 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID; Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-01** Sheet 1 of 1

OPL Title	**Nitrogen Inerting Before Steam-On**	**Discipline**	Process / Operations
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **nitrogen inerting before steam-on** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. Polymer powder + oxygen + heat can cause a dust explosion. The dryer must be fully inerted before any steam is admitted. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Start the N2 purge and raise FT-2302 flow above 200 kg/h	Permit valid & briefed
2	Confirm the vent O2 analyzer AT-2307 reads below 5 %	Parameter within limit
3	Only then admit steam gradually via TIC-2301 / TV-2301	Interlock status confirmed
4	Never bypass the FSLL-2302 low-flow inert trip to start faster	Change logged on DCS
5	If O2 rises above 8 %, the ASHH-2307 trip will stop steam and drive	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
Scheduled proof test of dryer outlet high-high temperature trip	Set-point within tolerance, minor zero drift corrected	Recalibrated TSHH-2301 to 125 degC and function tested to logic

6. KEY LEARNING POINTS

- Polymer powder + oxygen + heat can cause a dust explosion. The dryer must be fully inerted before any steam is admitted.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Thursday, 02 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-02** Sheet 1 of 1

OPL Title	**Gland Packing Replacement (PTFE 4526L)**	**Discipline**	Mechanical
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gland packing replacement (ptfe 4526l)** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. Worn gland packing lets powder and steam leak at the drum ends. This is the most frequent YD-2301 maintenance job and a housekeeping/safety issue. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate steam and N2, drain condensate, and LOTO the drive	Isolation & LOTO verified
2	Remove the old PTFE rings (8 pcs 4526L at discharge, 2 pcs 4505L at feed)	Parts match BOM / spec
3	Fit new rings one at a time, staggering the joints by 90 deg	Torque / clearance to spec
4	Tighten the gland follower evenly - snug only, never over-torque	No leak / smooth operation
5	Warm through slowly and re-check for leaks, re-nip after 24 h	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after .
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
SSLL-2305 speed low alarm, chain slack observed	Chain elongation with normal wear caused slip	Adjusted take-up, lubricated chain, checked sprocket wear

6. KEY LEARNING POINTS

- Worn gland packing lets powder and steam leak at the drum ends. This is the most frequent YD-2301 maintenance job and a housekeeping/safety issue.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Monday, 06 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-03** Sheet 1 of 1

OPL Title	**Chain & Sprocket Lubrication and Tension (SCM440)**	**Discipline**	Mechanical
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **chain & sprocket lubrication and tension (scm440)** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. A dry or slack drive chain slips and drops the drum speed, tripping SSLL-2305 and stopping drying. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Weekly apply the specified chain oil along the full drive chain	Isolation & LOTO verified
2	Inspect the SCM440 sprocket teeth for hooking/wear	Parts match BOM / spec
3	Check chain tension - mid-span sag should be 10-15 mm	Torque / clearance to spec
4	Adjust the take-up if sag is excessive and re-lube	No leak / smooth operation
5	Record chain elongation; replace the chain at 3 % stretch	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after .
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
SSLL-2305 speed low alarm, chain slack observed	Chain elongation with normal wear caused slip	Adjusted take-up, lubricated chain, checked sprocket wear

6. KEY LEARNING POINTS

- A dry or slack drive chain slips and drops the drum speed, tripping SSLL-2305 and stopping drying.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Friday, 10 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-04** Sheet 1 of 1

OPL Title	**Controlled Temperature Ramp-Up Procedure**	**Discipline**	Process / Operations
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **controlled temperature ramp-up procedure** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. Heating the drum too quickly warps the shell and cracks the steam tubes, causing tube leaks and long downtime. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Ramp TIC-2301 at a maximum of 10 degC per 10 minutes	Permit valid & briefed
2	Hold at 60 degC to let the drum warm through evenly	Parameter within limit
3	Watch drum thermal growth at the trunnion supports	Interlock status confirmed
4	Reach the 95-110 degC operating band slowly	Change logged on DCS
5	If uneven expansion / rubbing is heard, pause the ramp	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
Scheduled proof test of dryer outlet high-high temperature trip	Set-point within tolerance, minor zero drift corrected	Recalibrated TSHH-2301 to 125 degC and function tested to logic

6. KEY LEARNING POINTS

- Heating the drum too quickly warps the shell and cracks the steam tubes, causing tube leaks and long downtime.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Tuesday, 14 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-05** Sheet 1 of 1

OPL Title	**Rotary Joint Steam Leak Handling (Kadant)**	**Discipline**	Mechanical
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **rotary joint steam leak handling (kadant)** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. The Kadant rotary joint seal wears and leaks live steam at the drum head, a burn hazard and steam loss. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Identify the leak at the rotary joint carbon seal face	Isolation & LOTO verified
2	Reduce steam and plan an isolation window with operations	Parts match BOM / spec
3	Replace the rotary joint seal kit and inspect the nipple	Torque / clearance to spec
4	Check the O-ring (NBR) and the joint-to-shaft alignment	No leak / smooth operation
5	Leak test on steam-up before returning the dryer to service	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after .
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
SSLL-2305 speed low alarm, chain slack observed	Chain elongation with normal wear caused slip	Adjusted take-up, lubricated chain, checked sprocket wear

6. KEY LEARNING POINTS

- The Kadant rotary joint seal wears and leaks live steam at the drum head, a burn hazard and steam loss.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Saturday, 18 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-06** Sheet 1 of 1

OPL Title	**Outlet Moisture Control & Retention Time (MT-2306)**	**Discipline**	Process / Operations
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **outlet moisture control & retention time (mt-2306)** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. The product specification is below 300 ppm moisture. Rising outlet moisture means lost drying and off-spec powder. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the MT-2306 outlet moisture every shift	Permit valid & briefed
2	If moisture rises above 300 ppm, check steam pressure PT-2304 first	Parameter within limit
3	Verify drum speed and hence retention time is correct	Interlock status confirmed
4	Cross-check against the daily lab moisture sample	Change logged on DCS
5	Adjust feed rate or steam to bring moisture back in spec	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
Scheduled proof test of dryer outlet high-high temperature trip	Set-point within tolerance, minor zero drift corrected	Recalibrated TSHH-2301 to 125 degC and function tested to logic

6. KEY LEARNING POINTS

- The product specification is below 300 ppm moisture. Rising outlet moisture means lost drying and off-spec powder.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Wednesday, 22 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-YD-2301-07** Sheet 1 of 1

OPL Title	**Trunnion Bearing Inspection & Drum Float**	**Discipline**	Mechanical
Equipment	YD-2301 - POLYMER FLUID BED DRYER	**Area / Unit**	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	**P&ID; Ref**	TJC-LLD-PID-2301

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **trunnion bearing inspection & drum float** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. The support trunnions carry the full drum weight; worn rollers cause vibration and mis-tracking of the drum. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Do a listening / vibration check on both trunnions each round	Isolation & LOTO verified
2	Grease the trunnion bearings on the scheduled route	Parts match BOM / spec
3	Check the drum axial float stays within 5 mm	Torque / clearance to spec
4	Inspect the thrust roller for wear and correct tracking	No leak / smooth operation
5	Report any drum wobble or unusual thrust to reliability	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after .
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
SSLL-2305 speed low alarm, chain slack observed	Chain elongation with normal wear caused slip	Adjusted take-up, lubricated chain, checked sprocket wear

6. KEY LEARNING POINTS

- The support trunnions carry the full drum weight; worn rollers cause vibration and mis-tracking of the drum.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Sunday, 26 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-01** Sheet 1 of 1

OPL Title	**Reduction DC-3401A - 4-Step Overview**	**Discipline**	Process / Operations
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **reduction dc-3401a - 4-step overview** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. The catalyst reduction is done in four controlled steps: Inerting, Heating-Up, Reduction, then Cooling & Preservation. Skipping or rushing a step risks an H2/O2 explosion or ruined catalyst. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inerting: purge with N2 via FV-17343 until O2 below 100 ppm	Permit valid & briefed
2	Heating-Up: heater EA-3401 on, N2 at 275 kg/h, ramp TIC-17343 to 180-210 degC	Parameter within limit
3	Reduction: introduce H2, hold about 12 hours (PdO + H2 -> Pd)	Interlock status confirmed
4	Cooling: cool all bed below 29 degC, then hold under N2 at 2 kg/cm2	Change logged on DCS
5	Log every step transition and confirm permissives before the next step	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- The catalyst reduction is done in four controlled steps: Inerting, Heating-Up, Reduction, then Cooling & Preservation. Skipping or rushing a step risks an H2/O2 explosion or ruined catalyst.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Monday, 13 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-02** Sheet 1 of 1

OPL Title	**Inerting - O2 Freeing Below 100 ppm**	**Discipline**	Process / Operations
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **inerting - o2 freeing below 100 ppm** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. Hydrogen must never meet oxygen inside the reactor. Inerting is proven before any heating and again before hydrogen is admitted. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inject nitrogen through FV-17343 into the reactor	Permit valid & briefed
2	Purge until the outlet O2 analyzer AI-3401 reads below 100 ppm	Parameter within limit
3	Confirm all vent and drain paths are correctly lined up	Interlock status confirmed
4	Log the O2 value and time before proceeding to heat-up	Change logged on DCS
5	If O2 will not fall, check for air ingress at manways/flanges	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- Hydrogen must never meet oxygen inside the reactor. Inerting is proven before any heating and again before hydrogen is admitted.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Friday, 17 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-03** Sheet 1 of 1

OPL Title	**Heating-Up & Bed Homogenisation**	**Discipline**	Process / Operations
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **heating-up & bed homogenisation** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. The catalyst bed must be homogeneous at 120-135 degC before hydrogen is introduced, otherwise the reduction is uneven and can run away locally. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Energise heater EA-3401 (power on) after inerting is confirmed	Permit valid & briefed
2	Set the nitrogen carrier flow to about 275 kg/h	Parameter within limit
3	Raise TIC-17343 slowly to the 180-210 degC band	Interlock status confirmed
4	Monitor all TE-3401-1..8 until the bed is homogeneous 120-135 degC (~16 h)	Change logged on DCS
5	Do not admit H2 until every bed thermocouple is within 15 degC	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- The catalyst bed must be homogeneous at 120-135 degC before hydrogen is introduced, otherwise the reduction is uneven and can run away locally.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Tuesday, 21 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-04** Sheet 1 of 1

OPL Title	**Hydrogen Reduction Safety & Exotherm Control**	**Discipline**	Process / Operations
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **hydrogen reduction safety & exotherm control** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. Hydrogen is flammable over 4-75 % in air. During reduction the PdO->Pd reaction is exothermic and the bed temperature must be watched closely. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Introduce H2 only after inerting and bed homogeneity are confirmed	Permit valid & briefed
2	Hold a constant N2+H2 flow for about 12 hours	Parameter within limit
3	Watch TE-3401 for any exotherm approaching the TSHH-3401 limit	Interlock status confirmed
4	If bed exceeds 230 degC the logic auto-trips heater and blocks H2	Change logged on DCS
5	Keep the reactor vent path (PV-3404) available at all times	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- Hydrogen is flammable over 4-75 % in air. During reduction the PdO->Pd reaction is exothermic and the bed temperature must be watched closely.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Saturday, 25 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-05** Sheet 1 of 1

OPL Title	**Cooling-Down & Catalyst Preservation**	**Discipline**	Process / Operations
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **cooling-down & catalyst preservation** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. After reduction the catalyst is pyrophoric; it must be cooled slowly and kept under a nitrogen blanket to prevent re-oxidation or fire. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Decrease TIC-17343 slowly, no faster than 0.5 degC/s	Permit valid & briefed
2	Cool all bed thermocouples below 29 degC (ambient)	Parameter within limit
3	Switch off heater EA-3401 (power off) once cool	Interlock status confirmed
4	Isolate the reactor and hold nitrogen pressure at 2 kg/cm2	Change logged on DCS
5	Never open the reactor to air while the catalyst is active	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- After reduction the catalyst is pyrophoric; it must be cooled slowly and kept under a nitrogen blanket to prevent re-oxidation or fire.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Wednesday, 29 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-06** Sheet 1 of 1

OPL Title	**Multipoint Thermocouple TE-3401 Verification**	**Discipline**	Instrument
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **multipoint thermocouple te-3401 verification** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. The 8-point bed temperature profile drives the whole reduction; a failed or drifting thermocouple can hide a hot spot. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Verify all 8 TE-3401 points are reading and note the spread	Loop isolated / under permit
2	Flag any single TC more than 15 degC off its neighbours	Reference standard traceable
3	Cross-check the profile against TIC-17343 heater outlet	As-found & as-left recorded
4	Replace any open-circuit or drifting TC at the next opportunity	Trip / control action verified
5	Confirm TSHH-3401 selects the highest bed temperature for the trip	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer
N2 inert valve slow / sticking during inerting step	Actuator seals hardened, positioner out of calibration	Serviced actuator, replaced seals, recalibrated positioner

6. KEY LEARNING POINTS

- The 8-point bed temperature profile drives the whole reduction; a failed or drifting thermocouple can hide a hot spot.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Sunday, 03 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-DC-3401A-07** Sheet 1 of 1

OPL Title	**Manway Gasket & Box-Up After Catalyst Change**	**Discipline**	Mechanical
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	**Area / Unit**	3400 - CATALYST REDUCTION SYSTEM
Related Interlock	SEQ-3401 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-3401

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manway gasket & box-up after catalyst change** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. A leaking manway loses inert gas and risks air ingress that could ignite the fresh, active catalyst. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Always fit a new spiral-wound SS316/graphite gasket	Isolation & LOTO verified
2	Torque the manway bolts in a star pattern to the specified value	Parts match BOM / spec
3	Leak test the joint with nitrogen at 2 kg/cm2	Torque / clearance to spec
4	Record the catalyst batch number and the loaded volume (3.2 m3)	No leak / smooth operation
5	Re-instate PSV-3401 and all instruments before start-up	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled statutory PSV pop test	Set pressure drift +0.3 barg found	Overhauled PSV-3401, lapped seat, re-set to 6 barg, certified
End-of-life catalyst replacement at turnaround	Spent Pd/Al2O3 catalyst at end of run	Vacuumed out spent catalyst, inspected grids, loaded 3.2 m3 fresh cat.
Nitrogen leak at bottom manway during leak test	Spiral-wound gasket unevenly seated	Renewed SW SS316/graphite gasket, re-torqued in star pattern, leak tested.

6. KEY LEARNING POINTS

- A leaking manway loses inert gas and risks air ingress that could ignite the fresh, active catalyst.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Arya Wibisono (EMP-0912)	Thursday, 07 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-01** Sheet 1 of 1

OPL Title	**Lube Oil System Pre-Start & Priming**	**Discipline**	Mechanical
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **lube oil system pre-start & priming** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. With no lube oil pressure the crankshaft bearings fail in seconds; the pre-lube pump must build pressure before the machine is cranked. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Run the auxiliary lube oil pump and build pressure above 2 barg	Isolation & LOTO verified
2	Confirm PSLL-4504 shows healthy oil pressure	Parts match BOM / spec
3	Check the frame oil level and the filter differential pressure	Torque / clearance to spec
4	Verify no oil leaks and correct oil grade in the frame	No leak / smooth operation
5	Only then give the start command to the main motor	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- With no lube oil pressure the crankshaft bearings fail in seconds; the pre-lube pump must build pressure before the machine is cranked.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Friday, 24 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-02** Sheet 1 of 1

OPL Title	**Compressor Suction/Discharge Valve Inspection**	**Discipline**	Mechanical
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **compressor suction/discharge valve inspection** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Broken plate valves are the classic cause of high discharge temperature (TSHH-4503) and lost capacity on KC-4501. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate, depressurise and purge the cylinder with nitrogen	Isolation & LOTO verified
2	Remove the suction and discharge valve caps	Parts match BOM / spec
3	Inspect valve plates, springs and seats; replace as a complete set	Torque / clearance to spec
4	Check the valve gasket and seat for damage	No leak / smooth operation
5	Torque the valve caps to spec and leak test before start	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Broken plate valves are the classic cause of high discharge temperature (TSHH-4503) and lost capacity on KC-4501.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Tuesday, 28 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-03** Sheet 1 of 1

OPL Title	**Seal Gas (N2 Buffer) Management**	**Discipline**	Process / Operations
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **seal gas (n2 buffer) management** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. The nitrogen buffer keeps flammable process gas out of the crankcase; loss of seal gas is a fire/explosion risk. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Verify FT-4506 seal gas flow is established before start	Permit valid & briefed
2	Maintain the buffer differential above process pressure	Parameter within limit
3	Watch the distance-piece vent for signs of leakage	Interlock status confirmed
4	Respond immediately to a seal-gas low alarm	Change logged on DCS
5	Never run the compressor with seal gas lost	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- The nitrogen buffer keeps flammable process gas out of the crankcase; loss of seal gas is a fire/explosion risk.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Saturday, 02 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-04** Sheet 1 of 1

OPL Title	**Anti-Surge / Recycle Operation**	**Discipline**	Process / Operations
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **anti-surge / recycle operation** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. The recycle (spillback) valve protects the machine at low throughput and during start/stop; incorrect operation risks deadhead and rod overload. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Keep the anti-surge FV-4502 in auto during start-up	Permit valid & briefed
2	Avoid deadheading against a high discharge pressure (PSHH-4502)	Parameter within limit
3	Watch suction PSLL-4501 for gas starvation	Interlock status confirmed
4	Return the machine to load slowly, closing recycle gradually	Change logged on DCS
5	Investigate any repeated surge cycling with the vendor curve	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- The recycle (spillback) valve protects the machine at low throughput and during start/stop; incorrect operation risks deadhead and rod overload.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Wednesday, 06 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-05** Sheet 1 of 1

OPL Title	**Crosshead Vibration Monitoring (VSHH-4505)**	**Discipline**	Instrument
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **crosshead vibration monitoring (vshh-4505)** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. A rising crosshead vibration trend warns of loose running gear before it trips or fails. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trend VT-4501 crosshead vibration on the daily route	Loop isolated / under permit
2	Alarm at 8 mm/s and trip at 11 mm/s (VSHH-4505)	Reference standard traceable
3	Correlate any rise with TE-4507 main bearing temperature	As-found & as-left recorded
4	Investigate rod-load reversals and crosshead pin clearance	Trip / control action verified
5	Escalate a sudden step-change to reliability immediately	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Proof test of lube oil low-low trip	Set-point within tolerance	Recalibrated PSLL-4504 to 1.5 barg, function tested
Seal gas flow FT-4506 low, buffer pressure unstable	N2 buffer regulator diaphragm failed	Replaced seal gas regulator, re-established buffer dP
Proof test of KO drum high-high level trip	Float healthy, minor set-point trim	Verified LSHH-4510 float and trip action

6. KEY LEARNING POINTS

- A rising crosshead vibration trend warns of loose running gear before it trips or fails.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Sunday, 10 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-06** Sheet 1 of 1

OPL Title	**Suction KO Drum Draining (FA-4510)**	**Discipline**	Process / Operations
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **suction ko drum draining (fa-4510)** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Liquid carried over from the KO drum destroys valves and can bend piston rods; the boot must be drained regularly. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Drain the FA-4510 boot every shift on the operator round	Permit valid & briefed
2	Watch LT-4510 level and the LSHH-4510 high-high trip	Parameter within limit
3	Never run the compressor with a high KO drum level	Interlock status confirmed
4	Report frequent liquid carryover to process for investigation	Change logged on DCS
5	Confirm the auto-drain (if fitted) is functioning	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Liquid carried over from the KO drum destroys valves and can bend piston rods; the boot must be drained regularly.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Thursday, 14 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-07** Sheet 1 of 1

OPL Title	**Intercooler EA-4502 Performance Check**	**Discipline**	Mechanical
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **intercooler ea-4502 performance check** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Poor intercooling raises the 2nd-stage suction temperature and hence the final discharge temperature towards the trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the interstage temperature drop across EA-4502	Isolation & LOTO verified
2	Verify the jacket water flow FSL-4508 is adequate	Parts match BOM / spec
3	Clean the intercooler tubes if the approach temperature rises	Torque / clearance to spec
4	Log the interstage pressure and compare to datasheet	No leak / smooth operation
5	Report fouling trends so cleaning can be planned	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Poor intercooling raises the 2nd-stage suction temperature and hence the final discharge temperature towards the trip.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Monday, 18 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-01** Sheet 1 of 1

OPL Title	**Tube-Side Cleaning (Hydrojetting)**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **tube-side cleaning (hydrojetting)** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Hexane fouling on the tube side raises the differential pressure (PDT-5605) and cuts the heat duty, leaving cold solvent downstream. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and drain both sides, then purge with nitrogen	Isolation & LOTO verified
2	Open the channel cover and remove the old gasket	Parts match BOM / spec
3	Hydrojet the tube bores and inspect for pitting/erosion	Torque / clearance to spec
4	Verify tube-side dP restored and no residual blockage	No leak / smooth operation
5	Re-gasket with new SW SS316 gasket and box up to torque	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Hexane fouling on the tube side raises the differential pressure (PDT-5605) and cuts the heat duty, leaving cold solvent downstream.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Tuesday, 05 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-02** Sheet 1 of 1

OPL Title	**Gasket Replacement & Flange Box-Up**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gasket replacement & flange box-up** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Spiral-wound gaskets are single-use; reusing one causes hexane leaks at the channel and shell flanges. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Remove the old shell and channel gaskets completely	Isolation & LOTO verified
2	Clean and inspect the gasket seating faces for damage	Parts match BOM / spec
3	Fit new SW SS316/graphite gaskets, correctly centred	Torque / clearance to spec
4	Torque the flange bolts in a cross pattern in passes	No leak / smooth operation
5	Leak test on start-up and re-torque hot if permitted	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Spiral-wound gaskets are single-use; reusing one causes hexane leaks at the channel and shell flanges.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Saturday, 09 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-03** Sheet 1 of 1

OPL Title	**Steam Trap Health Check**	**Discipline**	Process / Operations
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **steam trap health check** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. A failed steam trap either floods the shell (killing duty) or blows live steam (wasting energy); regular checks catch both. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Compare the condensate temperature TI-5604 to steam temperature	Permit valid & briefed
2	Use ultrasonic or a listening stick to check trap operation	Parameter within limit
3	A cold trap means flooding; a continuously hot blow means leak-through	Interlock status confirmed
4	Check the condensate pot level gauge LG-5606	Change logged on DCS
5	Replace the trap element or trap as required	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- A failed steam trap either floods the shell (killing duty) or blows live steam (wasting energy); regular checks catch both.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Wednesday, 13 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-04** Sheet 1 of 1

OPL Title	**Thermal-Shock Avoidance on Start-Up**	**Discipline**	Process / Operations
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **thermal-shock avoidance on start-up** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Admitting steam onto a cold tube bundle too fast cracks the tube-to-tubesheet joints, causing internal leaks. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Establish solvent (tube-side) flow first	Permit valid & briefed
2	Admit steam slowly by ramping the TV-5602 / TIC-5602 set-point	Parameter within limit
3	Warm the exchanger through gradually	Interlock status confirmed
4	Avoid sudden large steps in the outlet temperature set-point	Change logged on DCS
5	Watch for knocking that indicates condensate hammer	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Admitting steam onto a cold tube bundle too fast cracks the tube-to-tubesheet joints, causing internal leaks.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Sunday, 17 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-05** Sheet 1 of 1

OPL Title	**Tube Leak Detection & Plugging**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **tube leak detection & plugging** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. A tube leak lets steam into the solvent (rising outlet temperature and water in the solvent); the leaking tube must be found and plugged. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and hydrotest the shell side to find the leak	Isolation & LOTO verified
2	Bubble-test to identify the specific leaking tube	Parts match BOM / spec
3	Plug both ends of the tube with tapered plugs	Torque / clearance to spec
4	Record the plugged-tube count (maximum 5 % of tubes)	No leak / smooth operation
5	Plan bundle replacement if the plugging limit is reached	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- A tube leak lets steam into the solvent (rising outlet temperature and water in the solvent); the leaking tube must be found and plugged.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Thursday, 21 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-06** Sheet 1 of 1

OPL Title	**Fouling Trend & Cleaning Schedule (PDT-5605)**	**Discipline**	Instrument
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fouling trend & cleaning schedule (pdt-5605)** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Trending the tube-side differential pressure predicts when cleaning is due before the duty is lost. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Log PDT-5605 differential pressure weekly	Loop isolated / under permit
2	Note the PDAH-5605 fouling alarm at 0.7 bar dP	Reference standard traceable
3	Correlate rising dP with any outlet-temperature shortfall	As-found & as-left recorded
4	Plan cleaning before the duty loss becomes critical	Trip / control action verified
5	Feed the trend into the exchanger cleaning program	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Outlet temperature hunting around set-point	Steam valve TV-5602 loop poorly tuned	Retuned TIC-5602 PID, verified stable control
Erratic tube dP reading	Impulse lines plugged with fines	Cleaned and re-zeroed PDT-5605 transmitter
Cracked gauge glass on condensate pot	Thermal/mechanical stress on glass	Replaced LG-5606 gauge glass and seals

6. KEY LEARNING POINTS

- Trending the tube-side differential pressure predicts when cleaning is due before the duty is lost.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Monday, 25 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-07** Sheet 1 of 1

OPL Title	**PSV-5607 Testing & Car-Seal Management**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **psv-5607 testing & car-seal management** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. The tube-side relief valve is the exchanger's overpressure protection and must be proven and never isolated in service. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm PSV-5607 is set to 16 barg on the nameplate	Isolation & LOTO verified
2	Carry out an off-line pop test per the statutory schedule	Parts match BOM / spec
3	Keep the PSV inlet/outlet isolation car-sealed open in service	Torque / clearance to spec
4	Record the test certificate and next due date	No leak / smooth operation
5	Never block-in a live PSV without an authorised spare	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- The tube-side relief valve is the exchanger's overpressure protection and must be proven and never isolated in service.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Friday, 29 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-01** Sheet 1 of 1

OPL Title	**Control Valve Stroke Calibration (DVC6200)**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **control valve stroke calibration (dvc6200)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Correct mapping of the 4-20 mA signal to 0-100 % valve stroke is essential for stable separator level control. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Connect ValveLink to the DVC6200 positioner	Loop isolated / under permit
2	Run the auto-calibration over the full stroke	Reference standard traceable
3	Verify 4 mA = closed and 20 mA = 38 mm fully open	As-found & as-left recorded
4	Check hysteresis and dead-band are within 1 %	Trip / control action verified
5	Save the calibration and record the as-left data	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Correct mapping of the 4-20 mA signal to 0-100 % valve stroke is essential for stable separator level control.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Saturday, 16 May 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-02** Sheet 1 of 1

OPL Title	**Instrument Air Supply & Filter Regulator**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **instrument air supply & filter regulator** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Dirty or low instrument air makes the valve sluggish or fail closed, upsetting FA-6710 level and tripping PSL-6702. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the PI-6702 air supply is at 1.4 barg	Loop isolated / under permit
2	Drain the filter regulator bowl of condensate	Reference standard traceable
3	Replace a clogged air filter element	As-found & as-left recorded
4	Confirm there are no air leaks on the tubing/fittings	Trip / control action verified
5	Verify PSL-6702 low-air alarm set-point	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Dirty or low instrument air makes the valve sluggish or fail closed, upsetting FA-6710 level and tripping PSL-6702.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Wednesday, 20 May 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-03** Sheet 1 of 1

OPL Title	**Packing Adjustment (PTFE V-Ring)**	**Discipline**	Mechanical
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **packing adjustment (ptfe v-ring)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Loose packing leaks hexane at the stem; over-tight packing makes the stem stick and the valve hunt. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate the line and depressurise before touching packing	Isolation & LOTO verified
2	Snug the packing nuts evenly in small increments	Parts match BOM / spec
3	Stroke the valve and check for a leak-free, smooth travel	Torque / clearance to spec
4	Do not over-torque - excessive friction degrades control	No leak / smooth operation
5	Record the packing adjustment in the valve history	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Manual handwheel jack on LV-6701 seized	Corrosion in handwheel thread	Overhauled handwheel assembly, greased thread
LV-6701 passing when closed (Class IV exceeded)	Seat/plug erosion from polymer bottoms	Lapped seat and plug, restored Class IV shut-off
Minor weep at body/bonnet joint	Spiral-wound body gasket relaxed	Renewed body gasket, torqued bonnet bolts

6. KEY LEARNING POINTS

- Loose packing leaks hexane at the stem; over-tight packing makes the stem stick and the valve hunt.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Sunday, 24 May 2026

P&ID

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-04** Sheet 1 of 1

OPL Title	**Positioner Response & Loop Check**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **positioner response & loop check** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Poor positioner tuning makes the level controller LIC-6701 hunt and the separator level cycle. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Step the LIC-6701 output and watch the valve response	Loop isolated / under permit
2	Check the ZSL/ZSH-6701 limit-switch feedback	Reference standard traceable
3	Tune the positioner gain for a stable, prompt travel	As-found & as-left recorded
4	Confirm there is no overshoot or stiction step	Trip / control action verified
5	Verify the valve seats fully at 4 mA (Class IV)	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Poor positioner tuning makes the level controller LIC-6701 hunt and the separator level cycle.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Thursday, 28 May 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-05** Sheet 1 of 1

OPL Title	**Manual Bypass (HV-6701) Operation**	**Discipline**	Process / Operations
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manual bypass (hv-6701) operation** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. The manual bypass lets operators keep the separator level safely while LV-6701 is isolated for maintenance. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Notify the panel and take LIC-6701 to manual	Permit valid & briefed
2	Crack open the bypass HV-6701 slowly	Parameter within limit
3	Throttle HV-6701 to hold FA-6710 level using LT-6710	Interlock status confirmed
4	Isolate LV-6701 (close block valves) for the work	Change logged on DCS
5	Reverse the steps and re-establish auto control afterwards	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 isolation for planned maintenance	Planned - bypass used to hold level	Lined up HV-6701 manual bypass, isolated LV-6701 safely
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Manual handwheel jack on LV-6701 seized	Corrosion in handwheel thread	Overhauled handwheel assembly, greased thread

6. KEY LEARNING POINTS

- The manual bypass lets operators keep the separator level safely while LV-6701 is isolated for maintenance.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Monday, 01 June 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-06** Sheet 1 of 1

OPL Title	**Fail-Closed & ESD Action Verification**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fail-closed & esd action verification** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. On air failure or ESD (XV-6701), the valve must drive closed to prevent gas blow-by from the separator. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trip the XV-6701 ESD solenoid during the test	Loop isolated / under permit
2	Confirm the actuator vents and the valve closes	Reference standard traceable
3	Time the full closure - it must be within 5 seconds	As-found & as-left recorded
4	Check the ZSL-6701 closed limit confirms on the DCS	Trip / control action verified
5	Reset the ESD and return the valve to auto	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- On air failure or ESD (XV-6701), the valve must drive closed to prevent gas blow-by from the separator.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Friday, 05 June 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-07** Sheet 1 of 1

OPL Title	**Actuator Diaphragm Inspection (Fisher 667)**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **actuator diaphragm inspection (fisher 667)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. A ruptured actuator diaphragm causes loss of control and a continuous air leak; periodic inspection prevents surprise failures. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inspect the Fisher 667 diaphragm for cracks or perishing	Loop isolated / under permit
2	Check the bench set is 0.2-1.0 barg	Reference standard traceable
3	Replace the NBR/fabric diaphragm if hardened	As-found & as-left recorded
4	Re-stroke and recalibrate after any diaphragm work	Trip / control action verified
5	Confirm the spring range still gives full 38 mm travel	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- A ruptured actuator diaphragm causes loss of control and a continuous air leak; periodic inspection prevents surprise failures.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Tuesday, 09 June 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-01** Sheet 1 of 1

OPL Title	**Barscreen Cleaning (Suction Screen)**	**Discipline**	Process / Operations
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **barscreen cleaning (suction screen)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A clogged barscreen starves the circulating water pumps and fouls the fill, reducing cooling for the whole unit. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Rake the SS304 barscreen every shift on the round	Permit valid & briefed
2	Remove leaves, debris and biological growth	Parameter within limit
3	Check the differential level across the screen	Interlock status confirmed
4	Dispose of debris correctly and log the cleaning	Change logged on DCS
5	Report rapid re-fouling to water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- A clogged barscreen starves the circulating water pumps and fouls the fill, reducing cooling for the whole unit.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Wednesday, 27 May 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-02** Sheet 1 of 1

OPL Title	**Gearbox Oil Level & Change (ISO VG 220)**	**Discipline**	Mechanical
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gearbox oil level & change (iso vg 220)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Low or degraded gearbox oil causes overheating (TSHH-7803) and gear failure high up on the fan deck. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the gearbox sight-glass oil level	Isolation & LOTO verified
2	Sample the oil for water contamination and metal	Parts match BOM / spec
3	Change the ISO VG 220 (4.5 L) per the schedule	Torque / clearance to spec
4	Verify the oil pressure PSL-7807 after refill	No leak / smooth operation
5	Record the oil change and next due date	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- Low or degraded gearbox oil causes overheating (TSHH-7803) and gear failure high up on the fan deck.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Sunday, 31 May 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-03** Sheet 1 of 1

OPL Title	**Fan Blade Inspection & Balance (FRP)**	**Discipline**	Mechanical
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fan blade inspection & balance (frp)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A cracked or eroded FRP blade unbalances the fan and causes the VSHH-7802 vibration trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Lock out the fan and inspect all 6 FRP blades	Isolation & LOTO verified
2	Check for cracks, leading-edge erosion and tip clearance	Parts match BOM / spec
3	Verify the blade pitch angle is uniform on all blades	Torque / clearance to spec
4	Re-balance the fan if vibration is high	No leak / smooth operation
5	Torque the blade root bolts to specification	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- A cracked or eroded FRP blade unbalances the fan and causes the VSHH-7802 vibration trip.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Thursday, 04 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-CT-7801-04** Sheet 1 of 1

OPL Title	**Circulating Water pH Control (AI-7805)**	**Discipline**	Process / Operations
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **circulating water ph control (ai-7805)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Wrong pH causes scaling or corrosion across the entire cooling water loop, not just this cell. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the AI-7805 pH each shift (target 7.0-8.5)	Permit valid & briefed
2	Check the acid / caustic dosing pumps are running	Parameter within limit
3	Cross-check against a manual lab titration	Interlock status confirmed
4	Adjust the dosing set-point as needed	Change logged on DCS
5	Report large pH swings to water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- Wrong pH causes scaling or corrosion across the entire cooling water loop, not just this cell.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Monday, 08 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-CT-7801-05** Sheet 1 of 1

OPL Title	**Drift Eliminator & Fill Inspection**	**Discipline**	Mechanical
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **drift eliminator & fill inspection** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Fouled or collapsed fill and drift eliminators cut the cooling performance and raise the basin temperature. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inspect the PVC fill for scaling and collapse	Isolation & LOTO verified
2	Check the drift eliminator packs are aligned	Parts match BOM / spec
3	Hose down fouled sections during a cell outage	Torque / clearance to spec
4	Report a thermal performance drop via TT-7801	No leak / smooth operation
5	Plan fill replacement if degradation is widespread	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- Fouled or collapsed fill and drift eliminators cut the cooling performance and raise the basin temperature.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Friday, 12 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-06** Sheet 1 of 1

OPL Title	**Gearbox Vibration Monitoring (VSHH-7802)**	**Discipline**	Instrument
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gearbox vibration monitoring (vshh-7802)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A rising vibration trend warns of gear or bearing wear before the fan trips or throws a blade. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trend the VS-7802 reading on the PdM route	Loop isolated / under permit
2	Alarm at 6 mm/s and trip at 9 mm/s (VSHH-7802)	Reference standard traceable
3	Correlate the rise with gearbox oil temperature TE-7803	As-found & as-left recorded
4	Inspect the floating drive-shaft couplings	Trip / control action verified
5	Escalate a step-change to reliability	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled pH analyzer calibration	Electrode drift	Recalibrated AI-7805 on buffer solutions, replaced electrode
Blowdown conductivity reading unreliable	Conductivity cell fouled with scale	Cleaned and recalibrated AI-7806 conductivity cell
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level

6. KEY LEARNING POINTS

- A rising vibration trend warns of gear or bearing wear before the fan trips or throws a blade.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Tuesday, 16 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-07** Sheet 1 of 1

OPL Title	**Conductivity / Blowdown Control (AI-7806)**	**Discipline**	Process / Operations
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **conductivity / blowdown control (ai-7806)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. High conductivity means the water is over-concentrated; automatic blowdown keeps the cycles of concentration in range. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the AI-7806 conductivity value	Permit valid & briefed
2	Verify the automatic blowdown valve XV-7806 operates	Parameter within limit
3	Maintain the target cycles of concentration	Interlock status confirmed
4	Log make-up versus blowdown volumes	Change logged on DCS
5	Coordinate chemical dosing with water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- High conductivity means the water is over-concentrated; automatic blowdown keeps the cycles of concentration in range.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Saturday, 20 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-01** Sheet 1 of 1

OPL Title	**Level Gauge / Bridle Verification (LT-8901)**	**Discipline**	Instrument
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **level gauge / bridle verification (lt-8901)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. A false level reading causes reflux pump cavitation (low) or column flooding (high); the transmitter must agree with the gauge. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Cross-check LT-8901 against the level bridle gauge	Loop isolated / under permit
2	Blow down the bridle to clear any plugging	Reference standard traceable
3	Confirm the LSHH-8901 and LSLL-8901 switch actions	As-found & as-left recorded
4	Verify the level tracks correctly through a level change	Trip / control action verified
5	Log any deviation and recalibrate if needed	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Blanket / vent split-range valve hunting	Split-range calibration overlap incorrect	Re-tuned PCV-8905 split-range, verified PT-8902 hold
Proof test of drum high-high level trip	Within tolerance	Recalibrated LSHH-8901 and verified EFF-1 column feed trip

6. KEY LEARNING POINTS

- A false level reading causes reflux pump cavitation (low) or column flooding (high); the transmitter must agree with the gauge.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Sunday, 07 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-FA-8901-02** Sheet 1 of 1

OPL Title	**PSV-8901 Testing & Car-Seal Management**	**Discipline**	Mechanical
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **psv-8901 testing & car-seal management** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The relief valve is the drum's last line of overpressure defence and must be proven and never isolated in service. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm PSV-8901 is set to 10 barg	Isolation & LOTO verified
2	Carry out the off-line pop test per the statutory schedule	Parts match BOM / spec
3	Keep the PSV isolation car-sealed open in service	Torque / clearance to spec
4	Tag and file the test certificate	No leak / smooth operation
5	Never block-in a live PSV without an authorised spare	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- The relief valve is the drum's last line of overpressure defence and must be proven and never isolated in service.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Thursday, 11 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-03** Sheet 1 of 1

OPL Title	**Water Boot Draining (LT-8903)**	**Discipline**	Process / Operations
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **water boot draining (lt-8903)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. Water accumulating in the boot upsets reflux quality and can slug the reflux pump; regular draining keeps the interface controlled. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Monitor the boot interface level LT-8903	Permit valid & briefed
2	Drain the boot N4 to the oily-water sewer via HV-8904	Parameter within limit
3	Stop draining before hydrocarbon breaks through (watch interface)	Interlock status confirmed
4	Log the drain volume each shift	Change logged on DCS
5	Report a rising water make to process	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904

6. KEY LEARNING POINTS

- Water accumulating in the boot upsets reflux quality and can slug the reflux pump; regular draining keeps the interface controlled.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Monday, 15 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-04** Sheet 1 of 1

OPL Title	**Nitrogen Blanketing (PCV-8905)**	**Discipline**	Process / Operations
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **nitrogen blanketing (pcv-8905)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The nitrogen blanket keeps air out of the hexane drum and holds a stable pressure; loss of blanket risks an explosive atmosphere. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Verify the PCV-8905 split-range holds PT-8902 pressure	Permit valid & briefed
2	Confirm the blanket makes up on falling pressure	Parameter within limit
3	Confirm the vent relieves on rising pressure	Interlock status confirmed
4	Respond immediately to a blanket-loss alarm	Change logged on DCS
5	Check the N2 supply header pressure is adequate	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904

6. KEY LEARNING POINTS

- The nitrogen blanket keeps air out of the hexane drum and holds a stable pressure; loss of blanket risks an explosive atmosphere.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Friday, 19 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-FA-8901-05** Sheet 1 of 1

OPL Title	**Reflux Pump Low-Level Protection (LSLL-8901)**	**Discipline**	Instrument
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **reflux pump low-level protection (lsll-8901)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. A low drum level cavitates and damages the GA-8920 reflux pump; the LSLL trip must be proven and never casually bypassed. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm the LSLL-8901 trip stops GA-8920	Loop isolated / under permit
2	Test the trip during commissioning and proof tests	Reference standard traceable
3	Never bypass the trip without an authorised permit	As-found & as-left recorded
4	Investigate any repeated low-level events	Trip / control action verified
5	Verify the pump restarts only after level recovers	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Blanket / vent split-range valve hunting	Split-range calibration overlap incorrect	Re-tuned PCV-8905 split-range, verified PT-8902 hold
Proof test of drum high-high level trip	Within tolerance	Recalibrated LSHH-8901 and verified EFF-1 column feed trip

6. KEY LEARNING POINTS

- A low drum level cavitates and damages the GA-8920 reflux pump; the LSLL trip must be proven and never casually bypassed.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Tuesday, 23 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-FA-8901-06** Sheet 1 of 1

OPL Title	**Corrosion / Thickness Inspection**	**Discipline**	Mechanical
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **corrosion / thickness inspection** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The boot and shell bottom see the water phase and are corrosion-prone; periodic thickness checks catch wall loss early. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Take UT thickness readings on the shell bottom and boot	Isolation & LOTO verified
2	Compare readings to the 3.0 mm corrosion allowance	Parts match BOM / spec
3	Inspect the mist pad and vortex breaker at turnaround	Torque / clearance to spec
4	Record the readings in the RBI / inspection file	No leak / smooth operation
5	Flag any accelerated wall loss to inspection engineering	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- The boot and shell bottom see the water phase and are corrosion-prone; periodic thickness checks catch wall loss early.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Saturday, 27 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-07** Sheet 1 of 1

OPL Title	**Manway Box-Up After Internal Inspection**	**Discipline**	Mechanical
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manway box-up after internal inspection** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. Correct gasket selection and bolt torque after inspection prevent a hexane leak from the manway. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Fit a new SW SS316/graphite manway gasket	Isolation & LOTO verified
2	Torque the manway bolts in a star pattern in passes	Parts match BOM / spec
3	Leak test with the nitrogen blanket	Torque / clearance to spec
4	Re-instate the PSV and all instruments	No leak / smooth operation
5	Confirm the drum is ready before re-introducing hydrocarbon	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- Correct gasket selection and bolt torque after inspection prevent a hexane leak from the manway.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Wednesday, 01 July 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.